
Chapter 2. Data, Measurements, and Data Preprocessing

Types of Data Sets: (1) Record Data

- Relational records
 - Relational tables, highly structured
- Data matrix, e.g., numerical matrix, crosstabs

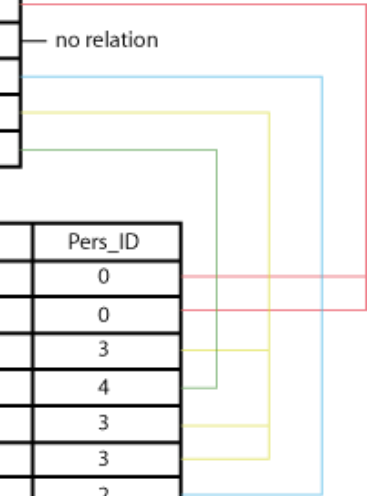
	China	England	France	Japan	USA	Total
Active Outdoors Crochet Glove		12.00	4.00	1.00	240.00	257.00
Active Outdoors Lycra Glove		10.00	6.00		323.00	339.00
InFlux Crochet Glove	3.00	6.00	8.00		132.00	149.00
InFlux Lycra Glove		2.00			143.00	145.00
Triumph Pro Helmet	3.00	1.00	7.00		333.00	344.00
Triumph Vertigo Helmet		3.00	22.00		474.00	499.00
Xtreme Adult Helmet	8.00	8.00	7.00	2.00	251.00	276.00
Xtreme Youth Helmet		1.00			76.00	77.00
Total	14.00	43.00	54.00	3.00	1,972.00	2,086.00

Person:

Pers_ID	Surname	First_Name	City
0	Miller	Paul	London
1	Ortega	Alvaro	Valencia
2	Huber	Urs	Zurich
3	Blanc	Gaston	Paris
4	Bertolini	Fabrizio	Rom

Car:

Car_ID	Model	Year	Value	Pers_ID
101	Bentley	1973	100000	0
102	Rolls Royce	1965	330000	0
103	Peugeot	1993	500	3
104	Ferrari	2005	150000	4
105	Renault	1998	2000	3
106	Renault	2001	7000	3
107	Smart	1999	2000	2



- Transaction data

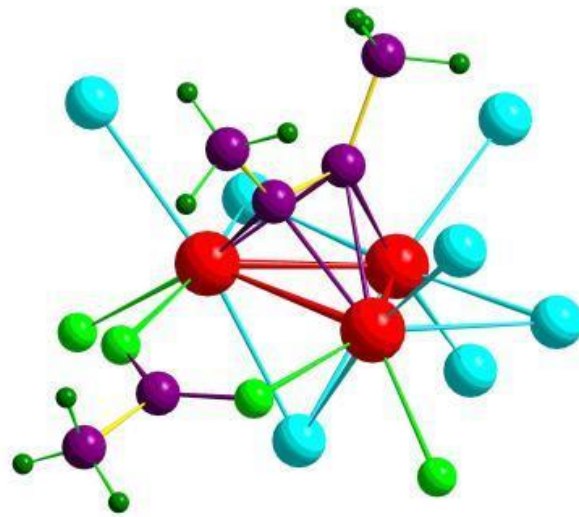
TID	Items
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

	team	coach	pla y	ball	score	game	wi n	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

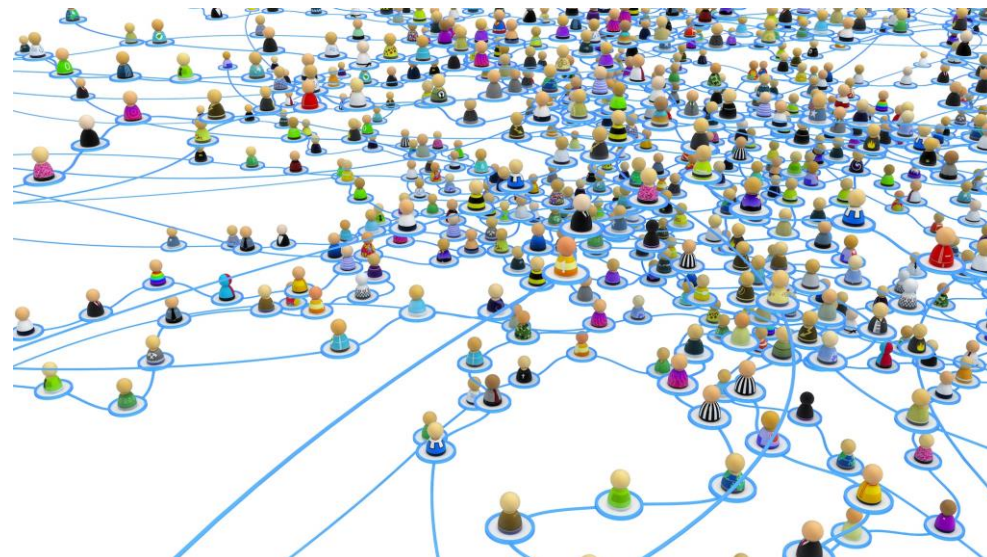
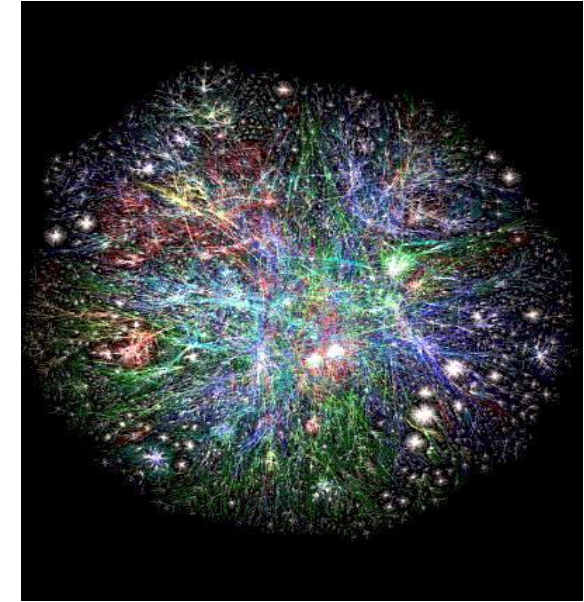
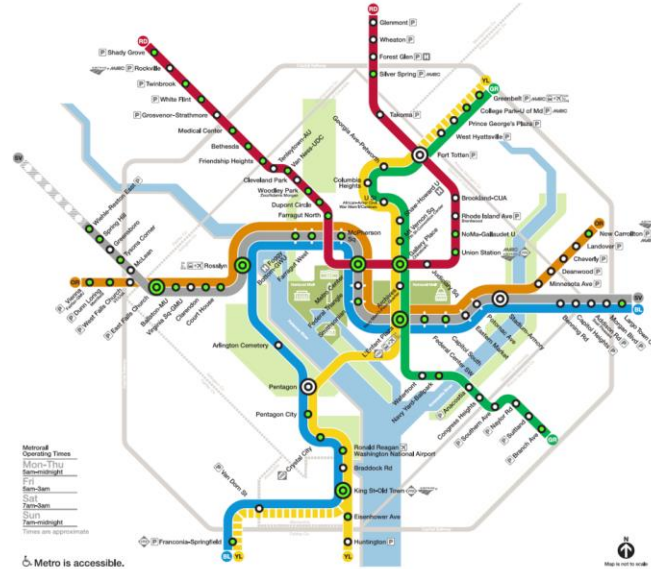
- Document data: Term-frequency vector (matrix) of text documents

Types of Data Sets: (2) Graphs and Networks

- ❑ Transportation network
- ❑ World Wide Web



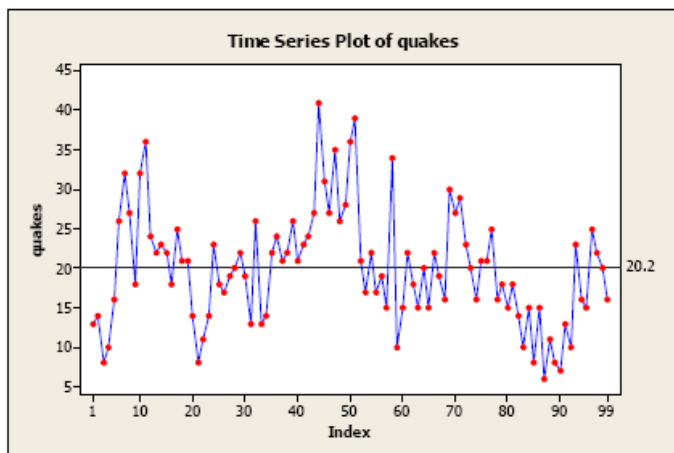
- ❑ Molecular Structures
- ❑ Social or information networks



Types of Data Sets: (3) Ordered Data

- Video data: sequence of images

- Temporal data: time-series



- Sequential Data: transaction sequences

- Genetic sequence data

Start

Human	GTTTGGAGG	---	ATGTTCAACAAATGCTCCTTTCATTCTCTATTTACAGACCTGCCGCA
Chimpanzee	GTTTGGAGG	---	ATGTTCAATAAATGCTGCTTTCACCTCTATTTACAGACCTGCCGCA
Macaque	GTTTGGAGG	---	ATGTTCAATAAATGCTCCTTTCATTCTCTATTTACAAACTGCCGCA
Human	GACAATTCTGCTAGCAGCC	TTTGTGCTATTATCTGTTTTCTAAACTTAGTAATTGAGTGT	
Chimpanzee	GACAATTCTGCTAGCAGCC	TTTGTGCTATTATCTGTTTTCTAAACTTAGTAATTGAGTGT	
Macaque	GACAATTCTGCTAGCAGCC	TTTGTGCTATTATCTGTTTTCTAAACTTAGTAATTGAGTGT	
Human	GATCTGGAGACTAA	-CTCTGAAATAAATAAGCTGATTATTTATTTATTTTCTCAAAACAA	
Chimpanzee	GATCTGGAGACTAA	CTCTGAAATAAATAAGCTGATTATTTATTTATTTTCTCAAAACAA	
Macaque	TATCTGGAGACTAA	CTCTGAAATAAATAAGCTGATTATTTATTTATTTTCTCAAAACAA	
Human	CAGAATACGATTTAGCAAA	TACTTCTTAAGATATTATTTTACATTTCTATATTCTCCTA	
Chimpanzee	CAGAATACGATTTAGCAAA	TACTTCTTAAGATATTATTTTACATTTCTATATTCTCCTA	
Macaque	CAGAATATGATTTAGCAAA	TACTTCTTAAGATATTATTTTGCATTTCTATATTCTCCTA	
Human	CCCTGAGTTGATGTGTGAGCA	AATATGTCACCTTTCATAAAGCCAGGTATACA	
Chimpanzee	CCCTGAGTTGATGTGTGAGCA	CGTATGTCACCTTTCATAAAGCCAGGTATACA	
Macaque	CCCTGAGTTGATGTGTGAGCA	AATATGTCACCTTTCATAAAGCCAGGTATATATACATTACG	
Human	GACAGGTAAGTAAAAA	ACATATTATTATTCTACGTTTTTGCCAAAAATTTTAAATTTT	
Chimpanzee	GACAGGTAAGTAAAAA	ACATATTATTATTCTACGTTTTTGCCAAAAATTTTAAATTTT	
Macaque	GACAGGTAAGTAAAAA	-CATATTATTATTCTAGGTTTTTGCCAAAGAGTTTAAATTTT	
Human	AACGTGTGCGCGTGTGTTGGTAA	---	
Chimpanzee	AACGTGTGCGCGTGTGTTGGTAA	---	
Macaque	AACGTGTGCGCGTGTGTTGGTAA	---	

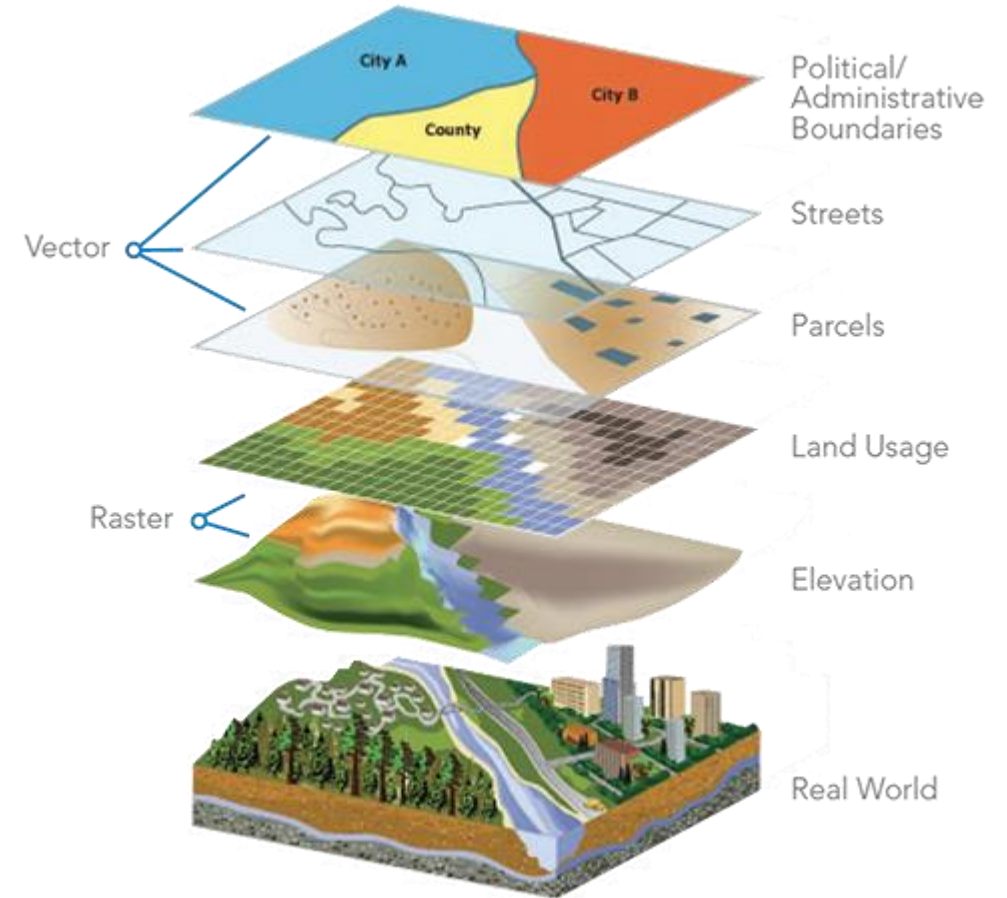
Types of Data Sets: (4) Spatial, image and multimedia Data

- Spatial data: maps



- Image data:

- Video data:



Data Objects

- ❑ Data sets are made up of data objects
- ❑ A **data object** represents an entity
- ❑ Examples:
 - ❑ sales database: customers, store items, sales
 - ❑ medical database: patients, treatments
 - ❑ university database: students, professors, courses
- ❑ Also called *samples, examples, instances, data points, objects, tuples*
- ❑ Data objects are described by **attributes**
- ❑ Database rows → data objects; columns → attributes

What is Data?

- ❑ Collection of data objects and their attributes
- ❑ An attribute is a property or characteristic of an object
 - ❑ Examples: eye color of a person, temperature, etc.
 - ❑ Attribute is also known as variable, field, characteristic, or feature
- ❑ A collection of attributes describe an object

Object is also known as record, point, case, sample, entity, or instance

Attributes

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Objects

Attributes

- **Attribute (or dimensions, features, variables)**

- A data field, representing a characteristic or feature of a data object.
- *E.g., customer_ID, name, address*

- **Types:**

- **Nominal** (e.g., red, blue)
- **Binary** (e.g., {true, false})
- **Ordinal** (e.g., {freshman, sophomore, junior, senior})
- **Numeric:** quantitative
 - Interval-scaled: 100°C is interval scales
 - Ratio-scaled: 100°K is ratio scaled since it is twice as high as 50°K
- Discrete vs. Continuous Attributes

Attribute Types

❑ **Nominal:** categories, states, or “names of things”

- ❑ *Hair_color* = {*auburn, black, blond, brown, grey, red, white*}
- ❑ marital status, occupation, ID numbers, zip codes

❑ **Binary**

- ❑ Nominal attribute with only 2 states (0 and 1)
- ❑ Symmetric binary: both outcomes equally important
 - ❑ e.g., gender
- ❑ Asymmetric binary: outcomes not equally important.
 - ❑ e.g., medical test (positive vs. negative)
 - ❑ Convention: assign 1 to most important outcome (e.g., HIV positive)

❑ **Ordinal**

- ❑ Values have a meaningful order (ranking) but magnitude between successive values is not known
- ❑ *Size* = {*small, medium, large*}, grades, army rankings

Numeric Attribute Types

- Quantity (integer or real-valued)

- **Interval**

- Measured on a scale of **equal-sized units**

- Values have order

- E.g., *temperature in C° or F°, calendar dates*

- No true zero-point

- **Ratio**

- Inherent **zero-point**

- We can speak of values as being an order of magnitude larger than the unit of measurement (**10 K° is twice as high as 5 K°**).

- e.g., *temperature in Kelvin, length, counts, monetary quantities*

Data Objects and Attribute Types

Numeric Attributes

- ❑ A **numeric attribute** is *quantitative*; that is, it is a measurable quantity, represented in integer or real values. Numeric attributes can be *interval-scaled* or *ratio-scaled*.
- ❑ **Interval-scaled attributes** are measured on a scale of equal-size units.
- ❑ The values of interval-scaled attributes have order and can be positive, 0, or negative. Thus, in addition to providing a ranking of values, such attributes allow us to compare and quantify the *difference* between values.
- ❑ Example A *temperature* attribute is interval-scaled.
- ❑ **Calendar dates** are another example. For instance, the years 2002 and 2010 are eight years apart.
- ❑ Temperatures in Celsius and Fahrenheit **do not have a true zero-point**, that is, neither 0C nor 0F indicates “no temperature.”

Data Objects and Attribute Types

- ❑ **Ratio-Scaled Attributes**

- ❑ A **ratio-scaled attribute** is a numeric attribute with an inherent **zero-point**.

- ❑ That is, if a measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value.

- ❑ **Example**

Has a true zero-point examples of ratio-scaled attributes include **count attributes** such as *years of experience* (e.g., the objects are employees)

Discrete vs. Continuous Attributes

□ Discrete Attribute

- Has only a finite or countably infinite set of values
 - E.g., zip codes, profession, or the set of words in a collection of documents
- Sometimes, represented as integer variables
- Note: Binary attributes are a special case of discrete attributes

□ Continuous Attribute

- Has real numbers as attribute values
 - E.g., temperature, height, or weight
- Practically, real values can only be measured and represented using a finite number of digits
- Continuous attributes are typically represented as floating-point variables

Data Quality, Data Cleaning and Data Integration

- ❑ Data Quality Measures
- ❑ Data Cleaning
- ❑ Data Integration
